Stochastic Optimization of Electric Vehicle Charging Infrastructure

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Abstract

The rapid growth of electric vehicle (EV) adoption has created increasing demand for efficient and economically viable charging infrastructure. This study develops a comprehensive stochastic optimization framework for the design of EV charging stations.

The system is modeled as a non-stationary $M(t)/G/c/c$ loss system, capturing time-varying customer arrivals and general service times. A data-driven approach is employed, combining Gaussian Mixture Models (GMM) for demand segmentation, Non-Homogeneous Poisson Processes (NHPP) for arrival modeling, and empirical distributions for service times.

A digital twin of the charging station is implemented using Discrete Event Simulation within a Monte Carlo framework, enabling the evaluation of system performance under uncertainty. An economic model is then constructed to translate technical performance into financial outcomes, with the objective of maximizing Net Annual Profit.

The optimization problem is solved by exhaustive search over a discrete set of charger configurations. Additionally, sensitivity analysis is conducted to assess the robustness of the optimal solution under parameter uncertainty.

The results demonstrate that a data-driven and simulation-based approach can effectively support decision-making in EV infrastructure planning, balancing service quality, operational efficiency, and long-term profitability.

0.1 Important Note: Model Limitations and Practical Considerations

The dataset used in this study originates from a real-world EV charging station; however, the actual system capacity (i.e., the number of installed chargers) is unknown. As a result, the observed arrivals correspond only to customers who were successfully served.

During peak hours, if the station operated with insufficient capacity, some customers may have left without charging and were therefore not recorded in the dataset. This implies that the observed demand may be slightly underestimated, particularly during high-demand periods.

Consequently, the proposed model is calibrated on partially censored demand data and may not fully capture the true underlying arrival process. For this reason, the results of this study should be interpreted as scenario-based economic analyses rather than absolute operational recommendations.

To address this limitation, an interactive decision-support tool (Streamlit dashboard) was developed. This tool allows decision-makers to explore "what-if" scenarios, such as increases in demand (e.g., +10% or +20%), and immediately observe the impact on both profitability and the optimal number of chargers.

This approach enhances the practical value of the model by enabling flexible, data-driven decision-making under uncertainty.

1 Introduction

The transition towards electric mobility has significantly increased the demand for reliable and efficient electric vehicle (EV) charging infrastructure. One of the central challenges in this context is determining the optimal number of chargers that ensures high service quality while maintaining economic viability.

EV charging stations operate under inherently stochastic conditions, characterized by uncertain customer arrivals, variable charging durations, and fluctuating demand patterns throughout the day. In addition, infrastructure investment decisions must account for both operational performance and financial outcomes, including installation costs, electricity prices, and lost revenue due to unserved customers.

In this study, we develop a comprehensive framework that integrates stochastic modeling, simulation, and economic analysis to address this problem. The system is modeled as a non-stationary loss system with time-dependent arrivals and general service times. To capture real-world variability, the methodology incorporates data-driven techniques, including clustering methods to distinguish between different demand regimes and probabilistic modeling of arrival processes.

A digital twin of the charging station is constructed using Discrete Event Simulation, allowing for the evaluation of system performance across a wide range of configurations. The simulation outputs are then translated into economic metrics through a detailed financial model, enabling the assessment of profitability.

The main objective is to determine the optimal number of chargers that maximizes Net Annual Profit. This is achieved through a discrete optimization approach, supported by

```
=== Data dimensions ===
(1965239, 4)

Columns:
['connectionTime_decimal', 'chargingDuration', 'kWhDelivered', 'dayIndicator']

First 5 rows
   connectionTime_decimal  chargingDuration  kWhDelivered  dayIndicator
0               15.332910          2.301191     11.469551             1
1               16.811495          4.941774      3.611695             1
2               21.151696          3.567173     14.455706             1
3               14.988554          3.089371      5.985330             1
4               15.078276          0.145047      1.274458             1
```

Figure 1: Dataset snippet

Monte Carlo simulation. Furthermore, sensitivity analysis is conducted to evaluate the robustness of the results under variations in key economic parameters.

The proposed framework provides a realistic and practical tool for infrastructure planning, combining mathematical rigor with real-world applicability.

2 Methodology and Model Design

The dataset consisted of four columns: *dayIndicator*, *connectionTime_decimal*, *chargingDuration*, and *kWhDelivered*. Each record corresponded to a single customer arrival.

From a theoretical perspective, the system can be modeled as an $M(t)/G/c/c$ loss system, characterized by a Non-Homogeneous Poisson arrival process, a general service time distribution, c parallel servers (chargers), and zero queue capacity. In such a system, when all chargers are occupied, any new arriving customer is blocked and lost. The main objective is to determine the optimal number of chargers c that maximizes the annual net profit. This objective accounts for installation and operational costs, electricity costs, and revenue losses due to blocked customers.

2.1 Data Analysis and Day Classification

The initial data exploration revealed significant heterogeneity across days. The average number of arrivals per day was approximately 66; however, some days had as few as 20 arrivals, while others exceeded 95. Using a single average value would therefore lead to oversimplification and inaccurate results.

```
mean = customers_per_day.mean()
var = customers_per_day.var()

print(mean, var)
```

```
66.39320945945946 598.409420423033
```

Figure 3: Mean and variance of number customers per day

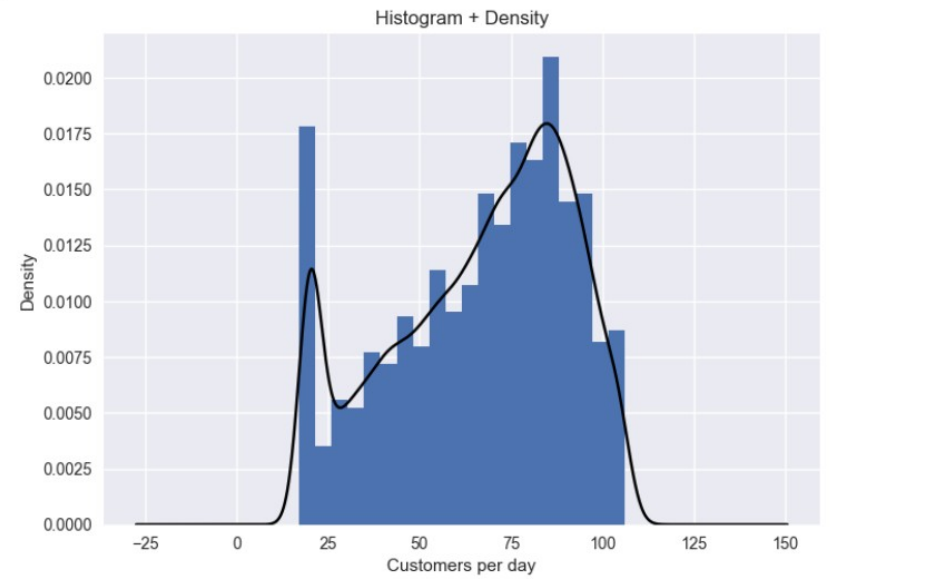


Figure 2: Probability of number of customers for a random day

To address this issue, a Gaussian Mixture Model (GMM) with two components was applied in order to identify two natural clusters of days: low-demand and high-demand. The GMM classified each day based on its total number of arrivals. For comparison purposes, a k-means clustering approach was also implemented, yielding similar but slightly different results. Ultimately, the GMM was selected, as it accounts for the variance within each cluster and provides a more realistic representation of the data.

Figure 4: Clustering results

```
=== GMM Clustering Results ===
type
High-demand    64.31
Low-demand     35.69
```

Figure 5: Probabilities for day type from GMM

```
=== KMeans Clustering Results ===
type_kmeans
High-demand    63.12
Low-demand     36.88
```

Figure 6: k-Means results for comparison with GMM

The clustering process produced the probabilities of occurrence for each day type (P_{Low}= 35.69 and P_{High}= 64.31), which were subsequently used in the stochastic simulation.

2.2 Arrival Rate Modeling

It was observed that the arrival rate is not constant throughout the day, with a clear peak occurring between 14:00 and 17:00.

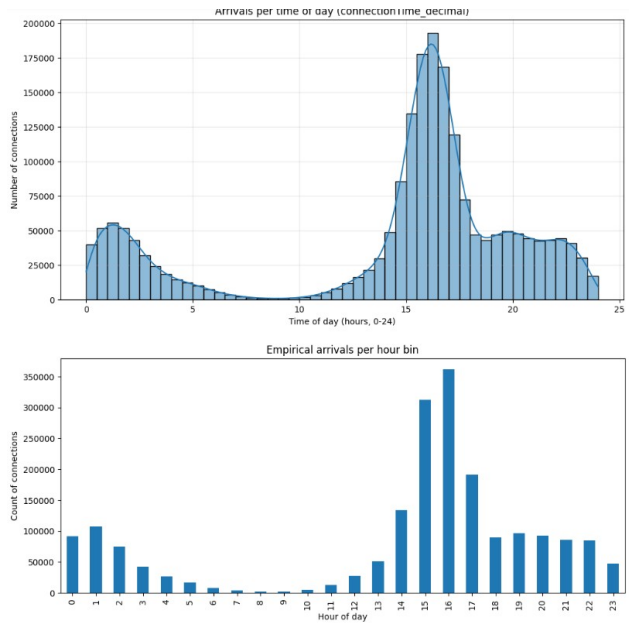
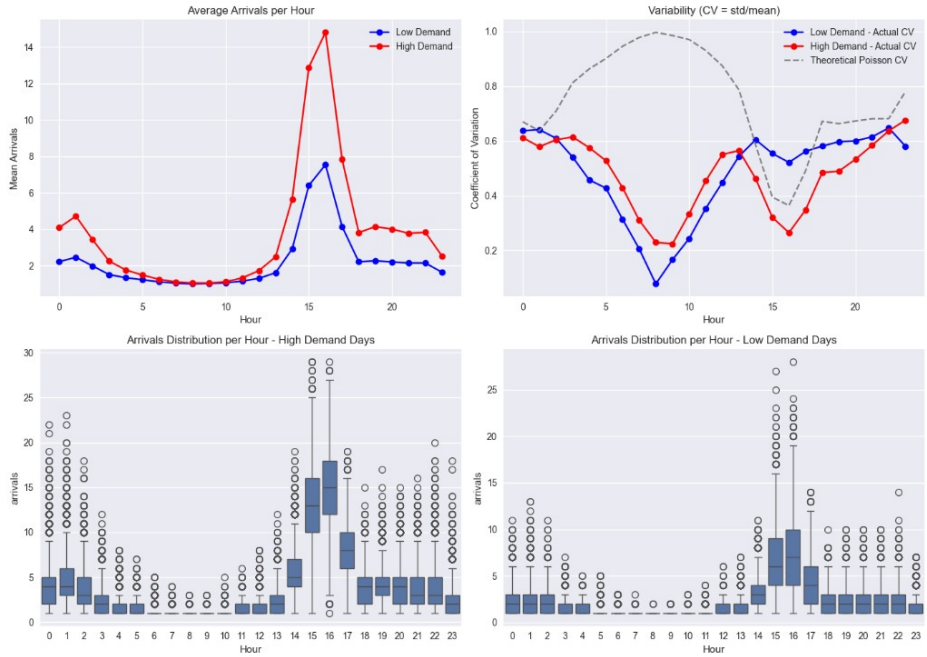


Figure 7: Distribution of arrivals per hour

Therefore, a Non-Homogeneous Poisson Process (NHPP) was adopted, where the arrival rate $\lambda(t)$ varies over time.

To validate the Poisson assumption, the coefficient of variation (CV = std/mean) was computed for each hour separately, for both low-demand and high-demand days. The results indicated relatively low variability (maximum CV approximately 0.65–0.68), which is even lower than what is typically expected for a Poisson process. This finding supports the validity of the modeling assumption.




```
============================================================
SUMMARY OF VARIABILITY
Max CV (Low) : 0.65  → Acceptable
Max CV (High): 0.68  → Acceptable
```

Figure 8: Hourly arrivals variance

```
HIGH DEMAND DAYS - Hourly Statistics
```

	hour	mean	std	count	min	max	cv	poisson_cv
0	0	4.094	2.506	18004	1	22	0.612	0.494
1	1	4.724	2.735	18367	1	23	0.579	0.460
2	2	3.433	2.074	17552	1	18	0.604	0.540
3	3	2.255	1.386	15057	1	12	0.615	0.666
4	4	1.768	1.017	12238	1	8	0.575	0.752
5	5	1.490	0.787	9226	1	7	0.528	0.819
6	6	1.253	0.538	5328	1	5	0.429	0.893
7	7	1.113	0.346	2565	1	4	0.311	0.948
8	8	1.058	0.243	1283	1	3	0.230	0.972
9	9	1.053	0.236	1423	1	3	0.224	0.975
10	10	1.122	0.374	3457	1	5	0.334	0.944
11	11	1.330	0.604	7805	1	6	0.454	0.867
12	12	1.733	0.954	12855	1	8	0.551	0.760
13	13	2.487	1.405	16484	1	12	0.565	0.634
14	14	5.657	2.615	18870	1	19	0.462	0.420
15	15	12.895	4.141	19036	1	29	0.321	0.278
16	16	14.815	3.924	19036	1	29	0.265	0.260
17	17	7.842	2.731	19026	1	19	0.348	0.357
18	18	3.822	1.853	18590	1	15	0.485	0.512
19	19	4.148	2.031	18635	1	17	0.490	0.491
20	20	4.011	2.138	18472	1	15	0.533	0.499
21	21	3.782	2.210	18141	1	16	0.584	0.514
22	22	3.834	2.442	17787	1	20	0.637	0.511
23	23	2.539	1.716	15092	1	18	0.676	0.628

Figure 9: Poisson for high

LOW DEMAND DAYS - Hourly Statistics

	hour	mean	std	count	min	max	cv	poisson_cv
0	0	2.227	1.419	7835	1	11	0.637	0.670
1	1	2.464	1.580	8312	1	13	0.641	0.637
2	2	1.991	1.214	7372	1	11	0.610	0.709
3	3	1.514	0.820	5311	1	7	0.541	0.813
4	4	1.342	0.614	3792	1	5	0.458	0.863
5	5	1.231	0.527	2694	1	5	0.428	0.901
6	6	1.118	0.351	1437	1	3	0.314	0.946
7	7	1.045	0.215	641	1	3	0.206	0.978
8	8	1.006	0.078	324	1	2	0.078	0.997
9	9	1.030	0.171	333	1	2	0.166	0.985
10	10	1.062	0.258	909	1	3	0.243	0.971
11	11	1.155	0.408	2230	1	4	0.354	0.931
12	12	1.308	0.587	4057	1	6	0.449	0.874
13	13	1.616	0.879	6126	1	7	0.544	0.787
14	14	2.923	1.768	9352	1	11	0.605	0.585
15	15	6.408	3.553	10409	1	27	0.554	0.395
16	16	7.548	3.938	10510	1	28	0.522	0.364
17	17	4.152	2.334	10110	1	14	0.562	0.491
18	18	2.217	1.288	8404	1	10	0.581	0.672
19	19	2.273	1.358	8459	1	10	0.597	0.663
20	20	2.208	1.325	8228	1	10	0.600	0.673
21	21	2.154	1.325	7765	1	10	0.615	0.681
22	22	2.152	1.393	7580	1	14	0.647	0.682
23	23	1.642	0.954	5539	1	7	0.581	0.780

Figure 10: Poisson for low

Event generation was performed using the thinning method, which preserves the fundamental properties of the NHPP, namely independent increments and Poisson-distributed arrivals within each subinterval.

2.3 Service Time Modeling

After testing several theoretical distributions (e.g., Exponential, Log-normal, Weibull), none provided a satisfactory fit to the data. As a result, an empirical distribution approach was adopted.

Specifically, for each arrival and each day type, the service time (charging duration) is randomly sampled from the observed *chargingDuration* values corresponding to the dataset.Although initial analysis suggested weak dependence between days, separate empirical distributions were retained for improved realism.

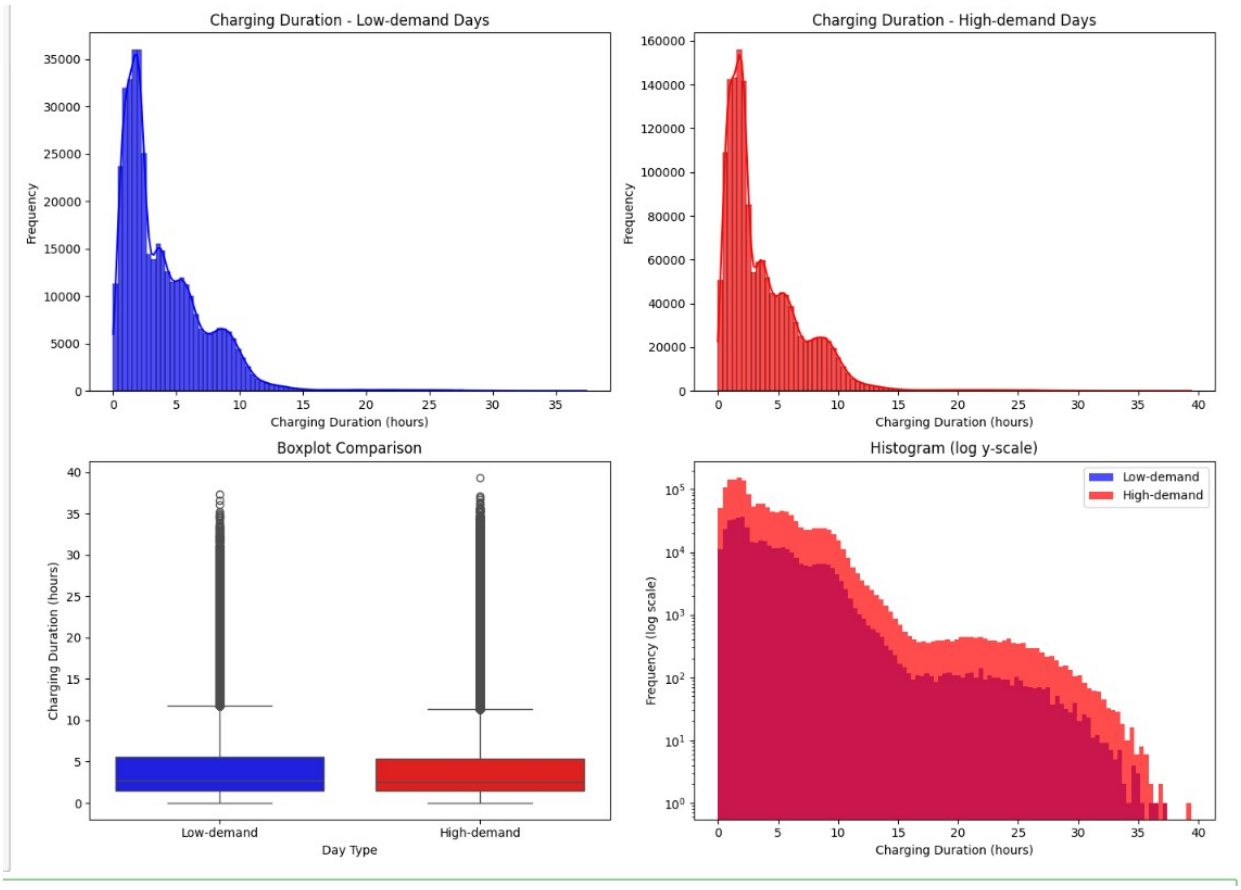


Figure 11: Charging duration for both days

3 Stochastic Simulation Implementation (Digital Twin)

To evaluate the performance of the charging station under different numbers of chargers, a digital twin of the real system was developed using Discrete Event Simulation (DES) within a Monte Carlo framework.

The simulation replicates thousands of synthetic operating days, enabling the extraction of statistically robust conclusions without interfering with the real system. For each simulated day, the following procedure is applied:

- **Day Type Selection:** Based on the probabilities P_{Low} and P_{High} obtained from the GMM clustering, the day is randomly classified as either low-demand or high-demand.
- **Arrival Generation:** Customer arrivals are generated using a Non-Homogeneous Poisson Process (NHPP) via the thinning method. This approach captures the temporal variability of the arrival rate within the day, including the peak period between 14:00 and 17:00.

```
=====================================================================
Λ(t) per hour(Low-demand days - GMM)
Hour : [ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23]
λ    : [2.227 2.464 1.991 1.514 1.342 1.231 1.118 1.045 1.006 1.03  1.062 1.155
 1.308 1.616 2.923 6.408 7.548 4.152 2.217 2.273 2.208 2.154 2.152 1.642]

=====================================================================
Λ(t) per hour(High-demand days - GMM)
Hour : [ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23]
λ    : [ 4.094  4.724  3.433  2.255  1.768  1.49   1.253  1.113  1.058  1.053
  1.122  1.33   1.733  2.487  5.657 12.895 14.815  7.842  3.822  4.148
  4.011  3.782  3.834  2.539]
```

Figure 12: Arrivals per hour per day type

- **Customer Service:** Each arriving customer is assigned:
  - A service time (charging duration) sampled from the empirical distribution.
  - An energy demand (kWh) sampled from the observed *kWhDelivered* distribution.

```
Real kWhDelivered samples available: 1,965,239
Mean = 9.44 kWh, Std = 5.70 kWh
Number of real kWh samples: 1,965,239
Mean   = 9.44 kWh
Std    = 5.70 kWh
Min    = 0.00 kWh
Max    = 62.54 kWh
```

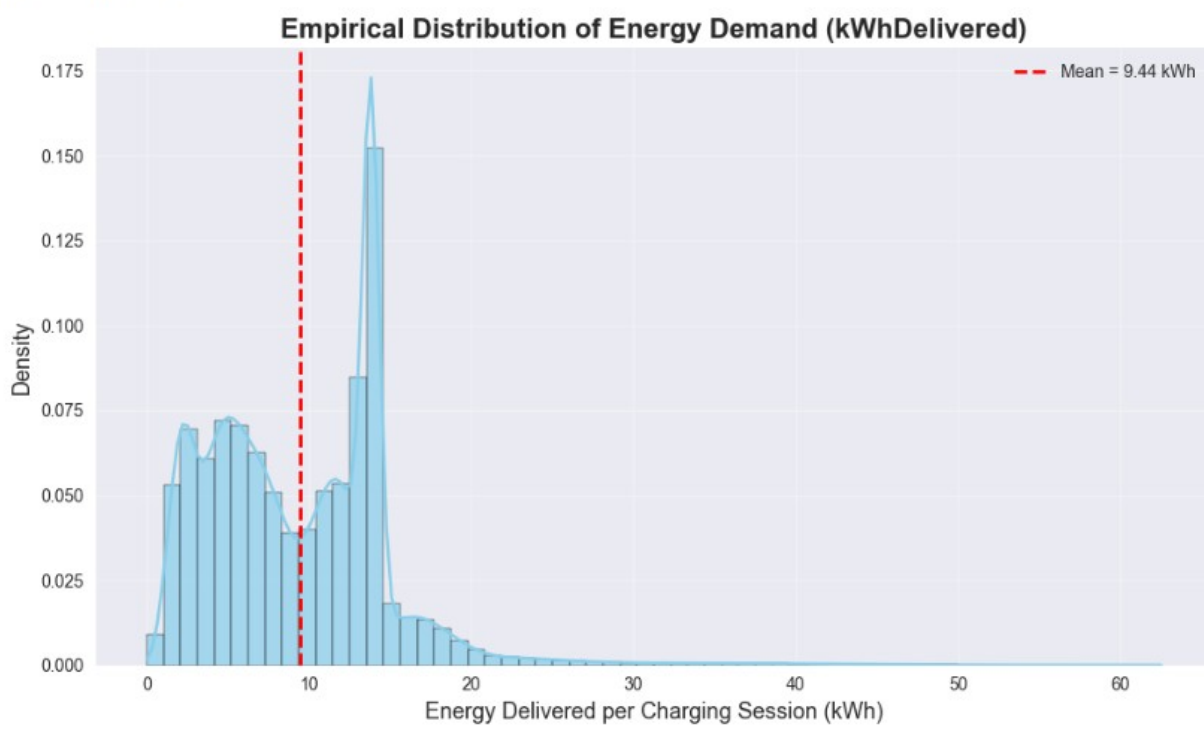


Figure 13: Distribution of kWh demand

- **System Operation:** The system is modeled as an $M(t)/G/c/c$ loss system. If a customer arrives while all chargers are occupied, the customer is blocked and lost. Event handling is implemented using a priority queue (heap structure) along with a *server_finish* array that tracks the completion time of each charger.

Through repeated simulation runs (typically 500–600 synthetic days per charger configuration), statistically reliable estimates of the system's key performance indicators are obtained:

- **Blocking Probability:** The proportion of customers that are rejected.
- **Utilization Rate:** The average utilization level of the chargers.
- **Total Energy Delivered (kWh):** The total energy supplied to customers.
- **Economic Metrics:** Revenue, operational costs, and net profit.

Annual scaling is performed by multiplying the results by the factor $\frac{365}{N_{\text{sim}}}$, where N_{sim} is the number of simulated days.

This approach enables not only the quantitative evaluation of each charger configuration, but also the exploration of system behavior under uncertainty. Such analysis would be difficult to achieve using only purely analytical methods due to the complexity of the $M(t)/G/c/c$ model.

4 Economic Model and Objective Function

Following the development of the stochastic digital twin, the next step was to translate the system's technical performance into economic terms. The ultimate objective is not merely to minimize the blocking probability or maximize charger utilization, but to maximize the *Net Annual Profit*.

To achieve this, a comprehensive economic model was developed, incorporating all major financial flows affecting the station's profitability. The model is based on annualized simulation outputs and includes both revenues and all relevant cost components.

The net annual profit is defined as:

$$\begin{aligned}\text{Net Annual Profit} = {} & \text{Annual Revenue} - \text{Annual Electricity Cost} \\ & - \text{Annual Maintenance Cost} - \text{Annual Lost Profit} \\ & - \text{Annualized Capital Cost}\end{aligned} \tag{1}$$

Each component is computed as follows:

- **Annual Revenue:** Calculated as the total annual energy delivered (in kWh) multiplied by the selling price per kWh. This term represents the direct income from EV charging services.
- **Annual Electricity Cost:** Computed as the same total energy multiplied by the electricity purchase price from the provider. This term is essential, as profit is derived from the margin between selling and purchasing prices.
- **Annual Maintenance Cost:** A linear cost per charger per year, directly proportional to the number of installed chargers c. This reflects the operational expenses associated with maintaining the infrastructure.
- **Annual Lost Profit:** Represents the economic impact of blocked customers. It is calculated as the annual number of rejected customers multiplied by a fixed profit margin per customer. This term captures the opportunity cost of unmet demand.
- **Annualized Capital Cost:** The annual depreciation of the initial installation investment. It is calculated by dividing the total installation cost ($c\times$ installation cost per charger) by the useful lifetime of the chargers (in years), thereby spreading the capital expenditure evenly over time.

The objective function is therefore the maximization of the Net Annual Profit as a function of the number of chargers c.

This formulation is particularly important, as it does not focus solely on technical performance (e.g., minimizing blocking probability), but instead achieves a balance between:

- the cost of additional chargers (capital and maintenance),
- the benefits of improved service levels (higher revenue and reduced lost profit),
- and the efficient utilization of the installed infrastructure (avoiding excessive idle capacity).

Through this economic framework, each configuration of c is evaluated both technically and financially, enabling the identification of the optimal investment that maximizes long-term station profitability.

5 Optimization Procedure

Following the development of the digital twin and the economic model, the problem is formulated as a discrete optimization problem. The objective is to determine the optimal number of chargers c that maximizes the Net Annual Profit.

5.1 Mathematical Formulation

The problem can be expressed as:

$$\max_{c \in \mathcal{C}} \quad f(c) = \mathbb{E}\left[\text{Net Annual Profit}(c)\right] \tag{2}$$

where:

- c is the number of chargers (discrete decision variable),
- $\mathcal{C} = \{20, 21, \ldots, 70\}$ is the search space (51 feasible values),
- $f(c)$ is the expected Net Annual Profit for a given c,
- $\mathbb{E}[\cdot]$ denotes the expectation, which cannot be computed analytically due to the stochastic nature of the system.

The objective function $f(c)$ is treated as a black-box function: it has no closed-form expression, no derivative information, and can only be evaluated through Monte Carlo simulation.

5.2 Intuitive Explanation

Intuitively, the goal is to determine how many chargers should be installed in order to maximize annual profit. Since evaluating all configurations in the real system is infeasible, a digital twin of the station is used.

For each candidate value of c (ranging from 20 to 70), the system is simulated multiple times. This allows us to estimate the average annual profit associated with each configuration. The optimal solution is then the value of c that yields the highest expected profit.

5.3 Optimization Method

Given that:

- c is a discrete variable,
- the objective function $f(c)$ is stochastic and lacks analytical structure,
- the search space is relatively small (51 possible values),

an exhaustive search (brute-force optimization) approach was adopted.

This method is both simple and highly reliable for small discrete problems, as it:

- guarantees identification of the global optimum within the defined range,
- does not rely on assumptions such as continuity or differentiability,
- enables direct comparison across all feasible configurations.

For each value of c, several hundred simulated days were executed to ensure a statistically stable estimate of $\mathbb{E}[\text{Net Annual Profit}(c)]$. The function $f(c)$ was then constructed, and the optimal value c^* was selected as:

$$c^* = \arg\max_{c \in \mathcal{C}} f(c) \tag{3}$$

5.4 Justification of the Approach

The choice of exhaustive search over more advanced optimization techniques (such as genetic algorithms, gradient-based methods, or surrogate models) is justified by the following:

- The search space is small (51 points), making the computational cost fully manageable.
- Accuracy is critical; exhaustive search guarantees identification of the true optimal solution within the examined range.
- The problem is stochastic, reducing the reliability of methods that depend on gradients or local approximations.

Thus, the optimization process is based on a rigorous mathematical formulation and systematic evaluation of all feasible solutions, ensuring both robustness and optimality of the final decision.

6 Sensitivity Analysis

After determining the optimal number of chargers c^*, it is essential to assess the robustness and stability of the solution under potential changes in key parameters. In practice, variables such as electricity prices, installation costs, and revenue rates may fluctuate over time. For this reason, a comprehensive sensitivity analysis was conducted.

6.1 Purpose of Sensitivity Analysis

The sensitivity analysis serves two main objectives:

- To evaluate how sensitive the optimal solution c^* and the Net Annual Profit are to variations in key economic parameters.
- To identify the parameters that have the greatest impact on profitability, enabling more effective risk management.

6.2 Methodological Approach

For each critical parameter p_i (e.g., revenue per kWh, electricity cost), three values were considered:

- Baseline value (p_i),
- Decreased value ($0.8 \times p_i$),
- Increased value ($1.2 \times p_i$).

For each scenario, the full optimization procedure was repeated, yielding a new optimal value of c and the corresponding Net Annual Profit.

The sensitivity of each parameter was quantified using the *Profit Swing*, defined as:

$$\text{Profit Swing}(p_i) = \max\big(f(c, p_i)\big) - \min\big(f(c, p_i)\big) \tag{4}$$

where $f(c, p_i)$ denotes the Net Annual Profit as a function of both the number of chargers and the parameter p_i.

6.3 Parameters Considered

The analysis focused on the most critical economic parameters:

- Revenue per kWh (selling price of electricity),
- Electricity Cost per kWh (procurement cost),
- Installation Cost per Charger,
- Lost Profit per Blocked Customer.

6.4 Results and Visualization

The results were presented using both a sensitivity table and a Tornado Chart. The Tornado Chart is a horizontal bar chart in which each bar represents the Profit Swing of a specific parameter. The longer the bar, the greater the impact of the parameter on the final profit.

This visualization enables immediate ranking of parameters according to their influence on profitability. Additionally, the stability of the optimal solution c^* was examined under different parameter scenarios.

6.5 Managerial Insights

The sensitivity analysis provides valuable managerial insights beyond purely technical evaluation. For example:

- If the *Revenue per kWh* exhibits the largest Profit Swing, then profitability is highly sensitive to pricing strategy.
- If the optimal c^* remains unchanged across a wide range of parameter values, the solution can be considered robust.

Overall, sensitivity analysis acts as a decision-support tool, enhancing the understanding of risks and identifying the key drivers that influence the long-term economic viability of the project.

7 Results and Discussion

7.1 The Bottom Line: What the Model Recommends

After simulating thousands of realistic operating days, the optimal configuration for the charging station is **34 chargers** under the given parameters (revenue per kWh = 0.75 €, electricity cost = 0.35 €, installation cost = 8,000 € per charger, maintenance cost = 800 € per year, and lost profit per blocked customer = 8 €),

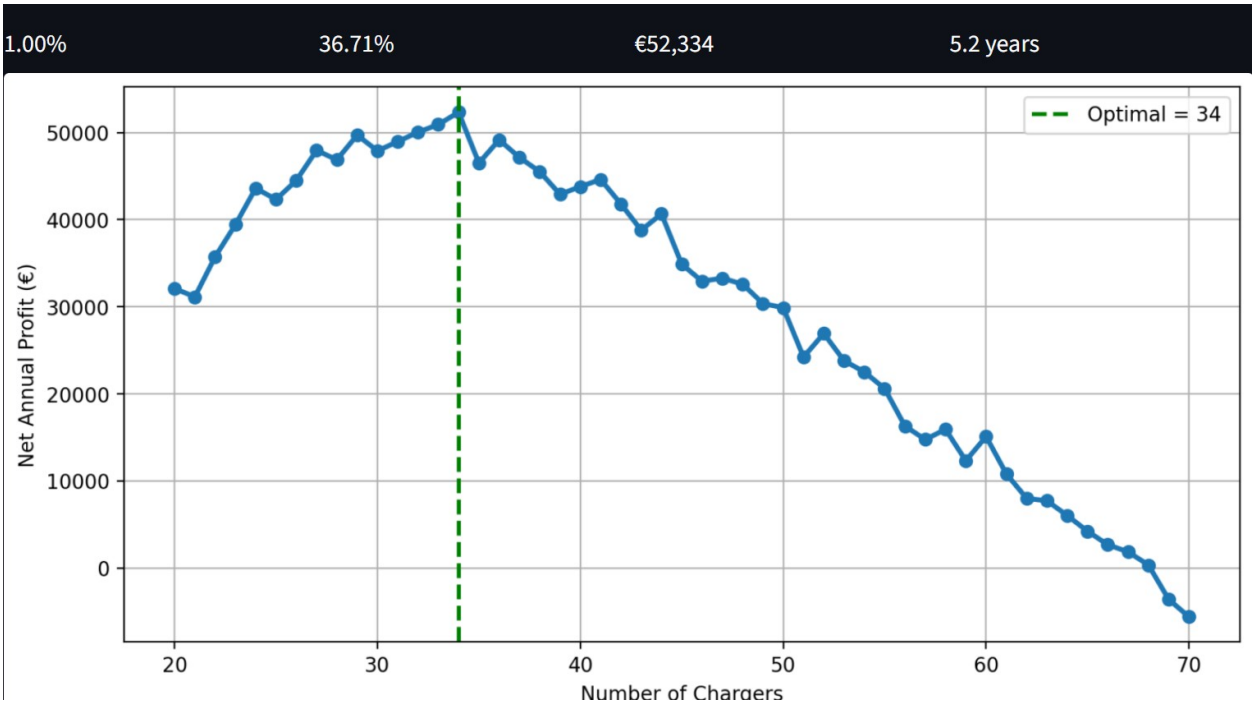


Figure 14: Optimal c for given parameters

At this setup, the station is expected to deliver:

KPI	Value	Business Meaning
Net Annual Profit	€52,334	Clear bottom-line contribution
Blocking Probability	1%	Almost zero lost customers
Charger Utilization	36.71%	Efficient use of capital
Payback Period	5.2 years	Fast return on investment

Table 1: Key Performance Indicators at the optimal configuration ($c^* = 34$).

Figure 14 shows the full profit curve. Profit rises sharply up to 34 chargers and then declines because the additional capital and maintenance costs outweigh the marginal revenue gains.

7.2 Why Exactly 34? The Business Trade-off Explained

The number 34 is not a random technical optimum; it represents the economic sweet spot where two opposing forces balance:

- **Too few chargers (e.g., 28–30):** Many customers are turned away during peak hours (14:00–17:00). Each blocked customer results in lost revenue and potential reputational damage.

- **Too many chargers (e.g., 40+):** High upfront installation costs (€8,000 per charger) and annual maintenance costs (€800 per charger) lead to underutilized assets.

At 34 chargers, the marginal revenue from serving an additional customer equals the marginal cost of installing and operating an additional charger. This corresponds to the maximum Net Annual Profit.

In practical terms:

34 chargers = maximum profit for the company

7.3 How Robust Is This Recommendation? (Sensitivity Analysis)

To assess robustness under uncertainty, the optimization was repeated by varying each key parameter by ±20%.

The main findings are summarized in Table 2.

Parameter	**Base**	**-20% Profit**	**+20% Profit**	**Swing (€)**	**Optimal** c
Revenue per kWh	0.750 €	€12,282	€94,289	€82,007	28–32
Electricity Cost	0.350 €	€71,100	€37,335	€33,765	30–36
Installation Cost	8,000 €	€59,347	€46,529	€12,818	28–32
Lost Profit per Block	8.00 €	€51,171	€52,792	€1,621	30–34

Table 2: Sensitivity analysis results (±20% variation in key parameters).

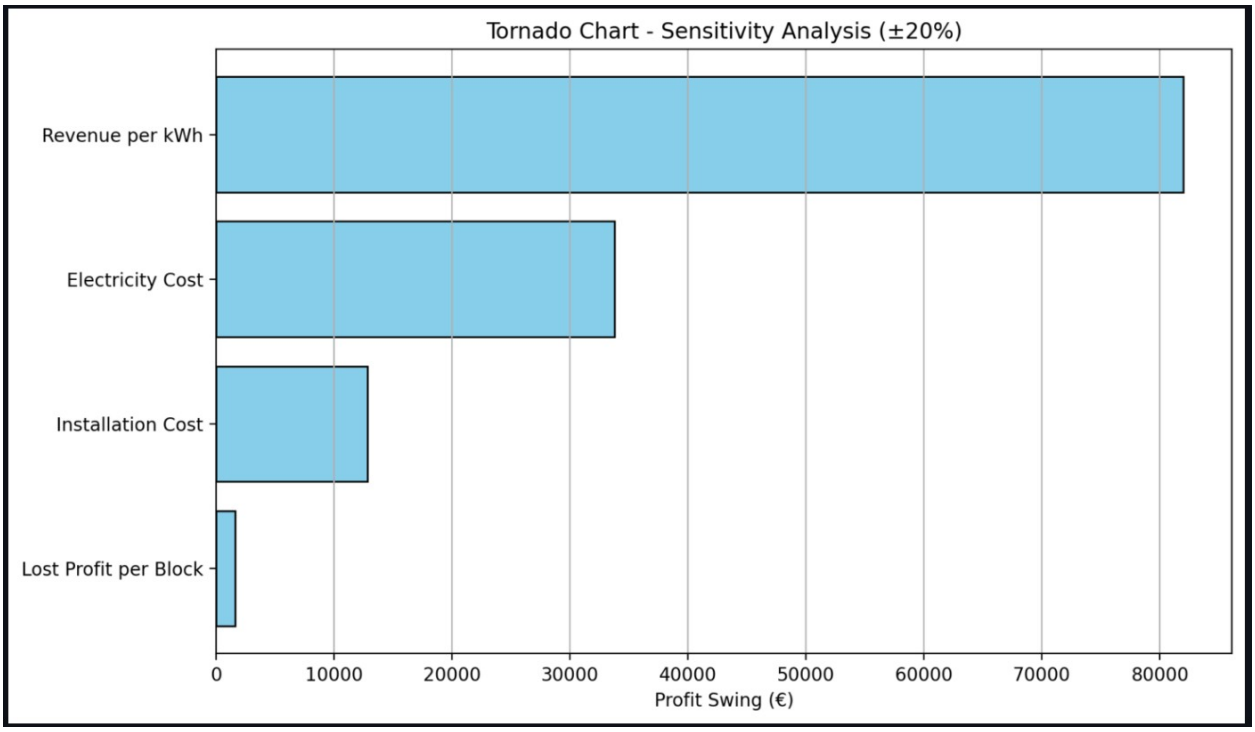


Figure 15: Impact of each parameter

As illustrated in Figure 15, the most influential parameter is the revenue per kWh, followed by electricity cost and installation cost. The lost profit per blocked customer has a significantly smaller impact.

Importantly, the optimal number of chargers remains relatively stable (between 28 and 36) across all scenarios, indicating strong robustness.

7.4 Managerial Takeaways and Business Value

This project delivers key value for decision-makers in the EV charging sector:

- A data-driven investment decision instead of heuristic estimation.
- A clear economic justification suitable for financial stakeholders.
- A flexible decision-support tool (e.g., Streamlit dashboard) allowing real-time scenario analysis.

Overall, the methodology transforms raw charging data into actionable, profit-oriented insights, balancing service quality and capital efficiency.

7.5 Conclusion

The proposed stochastic simulation and economic optimization framework successfully identifies the profit-maximizing charging infrastructure while remaining practical and robust.

By shifting the focus from purely technical metrics (e.g., blocking probability, utilization) to a Net Annual Profit objective, the analysis provides actionable guidance for real-world decision-making.

This work demonstrates that advanced data-driven methods, when translated into business-relevant insights, can generate measurable financial value in the rapidly growing EV sector.